

Automated Seizure Prediction

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ABSTRACT

In the past two decades, significant advances have been made on automated electroencephalogram (EEG)-based diagnosis of epilepsy and seizure detection. A number of innovative algorithms have been introduced that can aide in epilepsy diagnosis with a high degree of accuracy. In recent years the frontiers of computational epilepsy research have moved to seizure prediction, a more challenging problem. While anti-epileptic medication can result in complete seizure freedom in many patients with epilepsy, up to one-third of patients living with epilepsy will have medically-intractable epilepsy, where medications reduce seizure frequency but do not completely control seizures. If a seizure can be predicted prior to its clinical manifestation, then there is potential for abortive treatment to be given, either self-administered or via an implanted device administering medication or electrical stimulation. This will have a far-reaching impact on the treatment of epilepsy and patient's quality of life. This paper presents a state-of-the-art review of recent efforts and journal articles on seizure prediction. The technologies developed for epilepsy diagnosis and seizure detection are being adapted and extended for seizure prediction. The paper ends with some novel ideas for seizure prediction using the increasingly ubiquitous machine learning technology and in particular deep neural network machine learning.

Keywords – electroencephalogram; epilepsy; machine learning; seizure prediction.

1. INTRODUCTION

Seizures are due to the sudden change in the brain's electrical activity which results in alteration of awareness, often with associated involuntary abnormal movements [1, 2, 3]. Seizures occur as a result of hypersynchronous firing and propagation of abnormal neural networks [4, 5, 6]. Epilepsy is defined by having two or more unprovoked seizures or one unprovoked seizure with increased risk for future unprovoked seizures [7, 8, 9]. According to the World Health Organization (WHO), nearly 50 million people suffer from epilepsy [10].

The etiology of epilepsy can be categorized as structural, genetic-mediated, metabolic, or unknown [11]. Patients with epilepsy who experience loss of consciousness and convulsive movements are at risk for serious physical injury. Patients with uncontrolled generalized tonic-clonic seizures are also at increased risk of the fatal complication Sudden Unexplained Death in Epilepsy Patients (SUDEP) [12]. These patients are restricted from operating a motorized vehicle and unable to obtain meaningful employment due to uncontrolled seizures.

There is also strong evidence that people with long-standing uncontrolled epilepsy are at increased risk for permanent memory impairment, depression, anxiety, suicide, and other psychiatric disorders [13]. These factors negatively affect the quality of life of patients living with epilepsy. They also affect the lives of family and caretakers and represent an economic burden in terms of healthcare cost to their family members [14].

Following the seminal work of Adeli et al. (2003) [15], significant advances have been made on automated electroencephalogram (EEG)-based diagnosis of epilepsy and seizure detection in the past fifteen years [16, 17]. A number of innovative algorithms have been introduced that can perform the epilepsy diagnosis and seizure detection with a high degree of accuracy [18, 19, 20, 21]. Faust et al. [22] presented a review of wavelet-based techniques [23, 24, 25] for computer-aided seizure detection and epilepsy diagnosis. They conclude that a *“multi-paradigm approach through the adroit integration of wavelets, nonlinear dynamics and chaos theory, and neural networks advanced by Adeli and associates is the most effective method for automated EEG-based diagnosis of epilepsy.”*

In recent years the frontiers of computational epilepsy research have moved to seizure prediction, a more challenging problem. Therefore, the focus of this paper is a review of recent literature on the automated seizure prediction. Figure 1 depicts a block diagram of the proposed seizure prediction system employing the use of machine learning techniques for forecasting of

seizures by identifying the different episodes of seizure states in the EEG signals. In seizure prediction, the preictal EEG signal appearing a few minutes or a few hours before the seizure is predicted. In contrast, seizure detection is the classification of the EEG signals into seizure or non-seizure classes using the salient features in the EEG signals [26].

A computer-aided prediction (CAP) system is developed using signal processing and machine learning techniques [27, 28, 29]. It consists of pre-processing, features extraction, feature selection and ranking, and classification steps (see Figure 1). In this review, reviewed articles are divided into studies conducted using EEG data collected primarily from animals (rats and canine, summarized in Table 1) and humans (summarized in Table 2) subjects.



Figure 1 Block diagram for an automated seizure prediction system

2. SEIZURE PREDICTION USING ANIMALS

2.1 Signal Processing

Typically, the data collected are pre-processed for the removal of artifacts and noise before extraction of features. The power-line interference of 50 to 60 Hz is filtered to obtain a clean EEG signal for analysis.

2.2 Features Extraction

The EEG signals, in general, are highly chaotic with small signal magnitudes in millivolts. Usually, nonlinear feature extraction methods are preferred for feature extraction [26]. These nonlinear parameters can capture the hidden information and minute characteristics in the EEG signals effectively [26]. Nevertheless, some researchers still use the frequency and time-frequency domain feature extraction methodologies for development of a CAP. The features employed by researchers in Table 1 are discussed in this section briefly.

A) Nonlinear features

The largest Lyapunov exponent (LLE) [30, 31] parameter is an estimation of the chaos level of the EEG data. It can distinguish the different chaotic episodes present in the EEG data.

B) Frequency domain features

The spectral power analysis [32] is used to evaluate the frequency in each power spectrum. This technique is useful in investigating the subtle changes due to seizures in the EEG signals. The short-time Fourier transform [33] was also used to design a CAP system with the EEG data collected from 23 patients with intractable seizures.

C) Time-Frequency domain features

The wavelet transform is usually applied to decompose the EEG signals into its respective frequency components to represent the signal information as a function of time. Typically, in wavelet transforms, a mother wavelet is first selected depending on the nature of the signal. Then, the EEG signals are decomposed according to the morphology of the mother wavelet. The wavelet transformation is a useful technique to analyze the components of a time-varying and chaotic signal such as the EEG signal.

2.3 Feature Selection and Ranking

The aim in this step is to eliminate redundant features and select only characteristic features for classification. The principal component analysis (PCA) [34] and partial least squares (PLS) [35] have been applied to remove unwanted EEG signals that are not distinctive.

2.4 Classification

The classification process is the last building block of a CAP system. The prediction is made at this stage with the selected features fed into a classifier. From Table 1, the support vector machine (SVM) [36], artificial neural network (ANN) [37], and random forest (RF) [38, 39], classifiers have been employed to forecast the probability of seizure occurrence. Varatharajah et al. [40] present seizure prediction in canine epilepsy using three aforementioned machine learning approaches.

2.5 Regularisation

Due to the presence of the noise in the EEG signal, the features extracted will not be accurate and hence the classification may not be accurate. Hence, the classification may get affected due to the presence of noise in the EEG. Regularisation is the technique performed to eliminate the influence of noise on the classification and avoid the false positives. In seizure prediction the regularisation method proposed in [41] is being used to reduce the false positives and improve the classification accuracy.

2.6 Decision Making

After the elimination of noise, the output of regularisation is fed to the decision making function which is an alarm to give the output as preictal EEG if the seizure is predicted correctly.

Table 1: Summary of works conducted on automated seizure prediction using animal subjects.

Authors	Year	Objective of Study	Database/Methodology	Time of Prediction	Performance
Ouyang et al. [42]	2007	To develop a novel wavelet-based nonlinear similarity index to predict seizures	Rat EEG recordings Freiburg database - Wavelet and nonlinear dynamics measures	2.24 mins (Rats) 6.98 ± 1.50 mins (Freiburg database)	Wavelet approach is effective in predicting seizures
Rajdev et al. [43]	2010	To propose a real-time automated seizure prediction system developed from rats	Rat EEG recordings - Computer-aided prediction algorithm - Adaptive Wiener prediction technique	6.7 secs	Sensitivity: 92%
Varatharajah et al. [40]	2017	To investigate the possibilities of a seizure prediction using machine learning techniques	Canine EEG recordings - Computer-aided prediction algorithm → Power in band features → Time domain → Spectral coherence → PCA → PLS → SVM, ANN, RF classifier	1 min	Machine learning techniques portrayed possible seizure prediction capabilities
Shafique et al. [31]	2018	To discuss the process of developing	Rat EEG recordings - LLE - The Wolf's algorithm	10 secs	The use of Lyapunov exponents may

		nonlinear parameters with the help of Big data to predict seizures	• The Rosenstein-Kantz algorithm • Big data		refine the dependency of seizure prediction algorithm
Truong et al. [33]	2018	• To design a seizure prediction model using CNN	• Canine EEG recordings • Freiburg and CHB-MIT database • Short-time Fourier transform • CNN	• 30 secs	• Sensitivity: 75%

3. SEIZURE PREDICTION USING HUMAN SUBJECTS

3.1 Signal Processing

Similar to animal studies, signals collected from human subjects need to be pre-processed first before the feature extraction process.

3.2 Features Extraction

Various feature extraction techniques employed by researchers and summarized in Table 2 are reviewed in this section.

A) Statistical features

Statistical parameters such as mean, variance, skewness, and kurtosis have been used to extract information on any changes in the distribution and amplitude of the EEG data [44]. The common spatial pattern (CSP) [45] is another statistical approach used to separate the EEG data into subcomponents to yield a maximum distinction between the preictal and seizure EEG states. Phase correlation [46, 47] is another technique adopted to forecast the patterns behavior between the preictal and seizure episodes of the EEG data.

B) Nonlinear features

Correlation Dimension (CD) [48, 49] is often applied to EEG time-series data to measure the complexity of different episodes. Fractality Dimension (FD) is used to compare the rhythms in the EEG patterns and disclose the self-similarity present in the signals [50, 51]. Lempel-Ziv Complexity (LZC) measures the repeatability of episodic patterns in the EEG data [52, 49]. Hjorth

parameter can determine the statistical property of the EEG signals using three parameters namely activity, complexity, and mobility [53, 54].

The approximate entropy is a measure of repeatability of the EEG patterns [55, 56]. Hence, the more irregular the data will have a larger entropy value and vice versa. Spectral entropy also depicts the randomness of the EEG data. Similarly, wavelet entropy determines the level of disorder in the EEG data. It is able to detect the episodic patterns in the EEG signals. The wavelet energy discloses the uniformity of the EEG data in the different frequency bands [57, 58].

C) Frequency domain features

The spectral power analysis [32, 59] has been used to measure the frequency bands of the EEG data and an indication of seizure prediction. The fractional Fourier transform [60] is typically used to transform the EEG signals into frequency domain in order to extract characteristics features from different phase and magnitude components of the EEG data.

D) Time-Frequency domain features

In some research, a decomposition or transformation technique is applied to the pre-processed EEG data before extracting features. The bivariate empirical mode decomposition (BEMD) [61, 62] is employed to derive the intrinsic mode functions (IMFs) of the EEG data before extraction of features. Moreover, the features extracted from the wavelet transform such as discrete wavelet transform (DWT) [63] and continuous wavelet transform (CWT) [64] can be used for seizure prediction [63].

In DWT, the EEG signals are subjected to the low and high pass filters to decompose the data into approximate and detailed coefficients in various levels. Various hidden information can be extracted from these coefficients to study the nature of the signal. The CWT transforms the EEG signals into scalogram plots using dilation and translation of the wavelet functions. These coefficients are used to characterize different episodes of EEG signals. Different features can be extracted from this scalogram image to characterize the EEG signal.

Variational mode decomposition (VMD) [65, 66] and higher-order spectra (HOS) [67, 68] can also be combined with the existing nonlinear parameters to extract hidden characteristics from the EEG data in order to boost the performance of the CAP system. The VMD decomposes EEG signals into various frequency and amplitude modulated coefficients. The HOS is a nonlinear

parameter that consists of third and higher order moments and cumulants. It measures the nonlinear correlations among the various frequency components of the EEG data.

3.3 Features Selection / Ranking

Feature selection is carried using the following techniques: the PCA [34, 69], Gaussian mixture model (GMM) [70, 71], and minimum redundancy maximum relevance (MRMR) [72, 73].

3.4 Classification

According to Table 2, the following classifiers have been employed: linear discriminant analysis (LDA), SVM [36], RF [38], Bayesian [74], multi-layer perceptron (MLP) [75, 76], multilayer perceptron neural networks and radial basis functions neural networks [37], back-propagation neural network (BPNN) [77], probabilistic neural network (PNN) [78, 79, 80, 81]. Direito et al. [54] present seizure prediction using multi-channel high-dimensional feature sets and a multiclass SVM considering 216 patients from the European Epilepsy Database includes 185 patients with scalp EEGs and 31 with intracranial EEGs. Yuan et al. [82] developed a seizure prediction model using diffusion distance and Bayesian linear discriminate analysis with an open-source database that consists of 21 patients with 87 seizures.

3.5 Multi-Paradigm Approaches

Quyen et al. [83] tried to anticipate epileptic seizure through nonlinear analysis of similarity between EEG recordings using the phase space reconstruction technique to differentiate preictal and seizure states. Subsequently, a dynamic similarity index (DSI) was proposed to forecast seizures based on its numerical values [84, 85, 86].

Fujiwara et al. [87] conducted an interesting study using heart rate variability (HRV) signals to predict seizure. They reported that there are changes in the HRV signals during normal, preictal and seizure states. Therefore, a wearable device such as HeartSaver [88] can be used to monitor the heart rate of the seizure patient.

Yuan et al. [82] present an epileptic seizure prediction model using wavelet decomposition, diffusion distance, and a Bayesian linear discriminant classifier in intracranial EEGs obtained from 21 patients with 87 seizures.

Table 2: Summary of works conducted on automated seizure prediction using human subjects.

Authors	Year	Objective of Study	Database/Methodology	Time of Prediction	Performance
Petrosian et al. [89]	2000	To investigate the existence of a preictal stage before the seizure	Private data - Computer-aided prediction algorithm → Wavelet transform → RNN → Extended Kalman filter-based	2 mins	The proposed technique could detect the presence of preictal stages before an occurrence of a seizure
Winterhalder et al. [85]	2003	To develop a framework to assess the different seizure prediction techniques	Freiburg database - DSI	5 secs	The index can be employed as a characteristic to predict seizure based on a range of values
Aschenbrenner-Scheibe et al. [90]	2003	To evaluate the effectiveness of employing a nonlinear technique to predict seizure	Freiburg database - CD - Predictive dimension drops	Hippocampal group: 2.9 mins Neocortical group: 3.4 mins	Seizure prediction based on dimension drops are effective only to a certain extent
D'Alessandro et al. [51]	2003	To investigate the efficacy of choosing a set of optimal features for predicting seizures with hybrid feature selection technique	Private data - Computer-aided prediction algorithm → FD, Teager's algorithm, spectral entropy, second and third-level features → Genetic feature selection → PNN classifier	3.5 mins	- Sensitivity: 62.5% - Specificity: 90.47%
Maiwald et al. [86]	2004	To compare three nonlinear seizure prediction techniques	Freiburg database - DSI - CD - Increments of accumulated energy	5 secs	The dynamical similarity index achieved the best sensitivity rate out of the three nonlinear methods
Iasemidis et al. [91]	2005	To assess the performance of a real-time seizure prediction system	Private data - Chaos and angular frequency	91 min prior to seizure onset with an accuracy of 91.3%	Sensitivity: 81.82%
Mormann et al. [92]	2005	To compare the linear and nonlinear approaches to	Freiburg database - Univariate linear measures (second, third, and fourth statistical moments,	17 to 20.5 secs	A combination of univariate and bivariate measures could be a promising

		predicting seizures	the relative power of the bands, Hjorth parameters, CD, LLE) • Bivariate linear measure		approach to predict seizure
Mirowski et al. [93]	2009	• To investigate the efficacy of employing bivariate measures to predict seizures	• Freiberg database • Computer-aided prediction algorithm → Bivariate features → CNN	• 5 mins	• Sensitivity: 71%
Chisci et al. [94]	2010	• To propose an automated seizure prediction system	• Freiberg database • Computer-aided prediction algorithm → Autoregressive model → SVM classifier	• 2 secs	• Sensitivity: 100%
Feldwisch-Drentrup et al. [84]	2010	• To study how to enhance the performance of predicting seizures	• Freiburg database • Mean phase coherence • DSI	• -	• The integration of different techniques may yield better prediction performance
Park et al. [95]	2011	• To propose an automated seizure prediction system	• Freiburg database • Computer-aided prediction algorithm → Kalman filter → Kernel Fisher discriminant analysis → SVM classifier	• 10 secs	• Sensitivity: 97.5%
Arabi et al. [49]	2012	• To propose an automated seizure prediction system for a 30-minute and 50-minute horizon	• Freiberg database • Computer-aided prediction algorithm → CD, correlation entropy, noise level, LZC, and LLE → Rule-based approach	• 10 secs	30-min: • Sensitivity: 79.9% 50-min: • Sensitivity: 90.2%
Williamson et al. [69]	2012	• To propose an automated seizure prediction system	• Freiberg database • Computer-aided prediction algorithm → Eigenspectral of space-delay correlation and covariance matrices → PCA → SVM classifier	• 15 secs	• Correctly predicted 71 out of 83 seizures

Gadhoumi et al. [58]	2013	To propose an automated seizure prediction system	Private data - Computer-aided prediction algorithm → CWT → Wavelet energy and entropy → Fisher's test → Two-tailed Mann-Whitney test	10 mins	Sensitivity: < 85%
Li et al. [96]	2013	To propose a timely automated seizure prediction system	Freiburg database - Computer-aided prediction algorithm → Low-pass filter → Morphology filter → Spike rate detection	10 secs	Sensitivity: 75.8%
Zandi et al. [71]	2013	To propose an automated seizure prediction alarm system	Private data - Computer-aided prediction algorithm → Histogram → Variational GMM → Zero-crossing intervals	2 mins	Sensitivity: 88.34%
Zheng et al. [97]	2014	To propose an automated seizure prediction system	Freiburg database - Computer-aided prediction algorithm → BEMD → Mean phase coherence	20 secs	The proposed technique is able to pick up the synchrony changes before the occurrence of a seizure
Zhang et al. [56]	2014	To propose a low computation seizure prediction system for a 30-minute and 50-minute horizon	Freiburg database - Computer-aided prediction algorithm → Higuchi FD → Bayesian LDA → Kalman filtering	2 mins	30-min: - Sensitivity: 86.95% 50-min: - Sensitivity: 89.33%
Zhang et al. [98]	2014	To propose a real-time automated seizure prediction system	Private data - Computer-aided prediction algorithm - Approximate entropy	25 secs	Accuracy: 94.59%
Teixeira et al. [99]	2014	To propose an automated seizure prediction alarm system	EPILEPSIAE database - Computer-aided prediction algorithm → Auto-regressive modeling predictive error, decorrelation time, Hjorth	15.58 mins	Sensitivity: 73.55±24.83%

			- parameters, relative spectral power, spectral edge frequency and power, statistical moments, the energy of wavelet coefficients features → Kruskal-Wallis test → ANN classifier → SVM classifier		
Ghaderyan et al. [100]	2014	To propose an automated seizure prediction system and eliminate computational complexity	• Freiberg database • Computer-aided prediction algorithm → Univariate linear features → Fast Fourier transform → PCA → K-nearest neighbor-based undersampling → SVM classifier	• 20 secs	• Sensitivity: 100%
Bandarabadi et al. [59]	2015	To propose a statistical technique to select the preictal period for better prediction of seizure	• EPILEPSIAE database • Amplitude distribution histograms • Spectral power features	• 8 secs	• The proposed technique is able to provide predictive measures to predict seizures
Bandarabadi et al. [73]	2015	To propose an automated seizure prediction system	• EPILEPSIAE database • Computer-aided prediction algorithm → Relative spectral power features → MRMR feature selection → Amplitude distribution histogram → SVM classifier	• 5 secs	• Sensitivity: 75.8%
Parvez et al. [47]	2015	To propose an automated seizure prediction system	• Freiburg database • Computer-aided prediction algorithm → Phase correlation features → Least-square SVM classifier	• 5 mins	• Accuracy: 91.95%

Vahabi et al. [101]	2015	To propose a real-time automated seizure prediction system	Freiburg database - Bi-spectral analysis - Time-bi-frequency analysis	32 secs	Sensitivity: 100%
Behnam et al. [102]	2016	To propose an automated seizure prediction system	CHB-MIT database - Computer-aided prediction algorithm → Interpolated histogram feature → Seizure distribution model → Bayesian classifier → Hunting search algorithm → MLP classifier	6.64 secs	Accuracy: 86.56%
Fujiwara et al. [87]	2016	To propose a novel seizure prediction technique based on the analysis of heart rate variability	Private data (ECG and EEG data) - Analysis of heart rate variability signals - Time and frequency domain features were extracted from the signals - Multivariate statistical process control employed to statistically predict seizure	10 secs	- It is possible to use the heart rate to predict an epileptic seizure - Sensitivity: 91%
Zhang et al. [103]	2016	To propose a patient-specific automated seizure prediction system	Freiburg and CHB-MIT database - Computer-aided prediction algorithm → Spectral power → Kalman filter → SVM classifier	2 secs	Freiburg: - Sensitivity: 100% CHB-MIT: - Sensitivity: 98.68%
Sharif et al. [104]	2017	To propose an automated seizure prediction system	Freiburg database - Computer-aided prediction algorithm → Distribution of fuzzy rules on optimized Poincare samples → PCA → SVM classifier	12 mins	Sensitivity: 91.8-96.6%

Fei et al. [105]	2017	To propose an automated seizure prediction system	Private data and CHB-MIT database - Computer-aided prediction algorithm → Fractional Fourier transform → Modified LLE features → BPNN classifier	10 secs	- Accuracy: 89.67% - Sensitivity: 89.50% - Specificity: 89.75%
Direito et al. [54]	2017	To propose an automated seizure prediction system	EPILEPSIAE database - Computer-aided prediction algorithm → Auto-regressive modeling predictive error, decorrelation time, Hjorth parameters, relative spectral power, spectral edge frequency and power, statistical moments, the energy of wavelet coefficients features → SVM classifier	5 secs	Sensitivity: 38.47%
Chu et al. [106]	2017	To assess the capability of the attractor state analysis to forecast seizure	Private data and CHB-MIT database - Spectral feature - Fourier coefficients	20 secs	Sensitivity: 86.67%
Cho et al. [107]	2017	To study the efficiency of the phase lock value seizure prediction marker	CHB-MIT database - Computer-aided prediction algorithm → Bandpass filter → EMD → Multivariate EMD → Phase locking value → SVM classifier	1 sec	Accuracy: 83.17%
Zhang et al. [108]	2018	To develop a model for quantifying the dynamic evolution of EEG signals during seizures	Private data - A mathematical model based on a differential equation	10 secs	Synaptic plasticity has considerable influence on the period of seizure

Yuan et al. [82]	2018	To develop an automated seizure prediction algorithm using a novel feature	Freiburg database - Computer-aided prediction algorithm → Wavelet transform → Diffusion distance → Bayesian linear discriminant analysis	10 secs	30-min: - Sensitivity: 85.11% 50-min: - Sensitivity: 93.62%
			CHB-MIT database - Computer-aided prediction algorithm → Statistical features → Zero crossings → Wavelet transform → Power spectral → Cross-correlation → Graph theory → LSTM	15 mins to 120 mins	Sensitivity: < 99%

It can be seen from both Tables 1 and 2 that, various nonlinear features such as LLE, CD, entropies, FD, phase locking values, Teager energy and Poincare geometry have been used for seizure prediction. Wavelet transform, EMD, AR modelling, and higher order spectra methods have been employed. The SVM classifier is the most widely-used machine learning technique for the automated prediction of seizure. The most recent studies, however, employ deep learning techniques for prediction of seizure [33, 109].

4. CHALLENGES IN SEIZURE PREDICTION

First and foremost, longitudinal EEG data are required to verify the accuracies of a seizure prediction model. However, these longitudinal data are not easy to collect. Moreover, extracting minute details from the preictal EEG signals to predict seizure accurately is challenging. Additional research on feature extraction techniques is needed for development of an accurate seizure prediction model.

Second, the EEG signals are chaotic in nature. The presence of noise in the scalp EEG signals affects the accuracy of detecting preictal EEG states. This may result an increase in the percentage of false positives. Wavelet transform is often used for denoising in seizure prediction studies [22]. A more effective approach is wavelet packet transform where multiresolution is performed on both approximations and details of each EEG signal [110]. Further, a more sophisticated machine learning algorithm can be used for a more accurate prediction of seizure. This idea is expanded further in the next section.

Finally, it is difficult to achieve an accurate early seizure prediction due to the limited number and length of EEG data. Ideally, the seizure needs to be predicted early enough to allow for a suitable abortive intervention via anti-epileptic medication or neurostimulation before the development of clinical symptoms. Therefore, big data is necessary to overcome the shortage of training EEG data [16]. A graphics processing unit (GPU) may be needed to speed up the computation and reduce the training and testing times. Thus, authors advance the application of big data and deep learning algorithms for developing a fast and accurate CAP system. Tables 1 and 2 include the time required to predict seizure reported by various researchers which varies from 1 second to 120 minutes with limited subjects. These methods need to be tested with large databases before the clinical use.

5. FUTURE DEVELOPMENT

As noted in Table 2, Petrosian et al. [89] and Mirowski et al. [93] made use of the recurrent neural network (RNN) [111] and a five-layer convolutional neural networks (CNN) [112], respectively, to predict the seizure. Recently, Tsiouris et al. [109] designed a 2-layer long short-term memory (LSTM) [113] model to forecast seizure.

The CNN and LSTM are also known as deep learning [114, 115]. The term deep in deep learning refers to more than two layers of NN combined. Each NN consists of neurons connected from one layer to another. CNN is the most well-established deep learning technique. It has been employed in diverse applications such as medical image recognition [116, 117], physiological signals analyses [118], diagnosis of Alzheimer's [119], diagnosis of the Creutzfeldt -Jakob Disease [120], and health monitoring of structures [121, 122, 123]. The basic structure of a CNN comprises of a convolutional layer, a pooling layer, and a fully-connected layer (see Figure 2). In the convolutional layer, a set of learnable filters is convolved with the input EEG signals. During the convolution process, the network picks up distinctive features and produces a feature map

(output). The pooling layer, implemented after the convolution layer, aims to control overfitting of the network. Finally, the fully-connected layer connects all the neurons together to the final layer where the prediction is made. With CNN, significant features can be automatically learned directly from the EEG data and no hand-crafted features are required. Since the availability of longitudinal data is very limited, the model is able to learn adaptively from the training EEG signals to pick up preictal EEG episodes for the prediction of seizure.

LSTM has also been used in conjunction with CNN for the analysis of physiological signals [124, 125] as well as speech recognition and handwriting recognition [126]. LSTM is capable of learning long-term dependencies. Its architecture consists of a memory block with three different gates, namely, input, forget, and output [113]. These gates control and decide which information to be retained or discarded from the network structure.

The autoencoder is an unsupervised NN that is trained on unlabeled input data learns the representation (encoding) of these data and then decodes (reconstructs) the data as precisely as possible [127]. Deep belief network (DBN) and restricted Boltzmann machine (RBM) are two typical autoencoder systems [127]. The DBN is made up of stacked layers of the RBM network. Just like the autoencoder, an RBM network trains the network to reconstruct the input data through forward and backward translations [128].

Recently, Acharya et al. [16] designed a *thirteen*-layer CNN to differentiate normal, preictal, and seizure states. Their proposed model achieved sensitivity and specificity of 90% and 95%, respectively with minimal supervision. The research team is currently extending that research to forecast seizure attacks.

Advantages of deep machine learning techniques are robustness against noise present in the EEG signals and no need for a feature extraction step. Deep learning models can self-learn from the input training data [118]. On the other hand, they need a lot of data to learn. Big data is necessary to train the algorithm to achieve high sensitivity performance in predicting the seizure. Training of the algorithm is also computationally more expensive compared to the tradition machine learning techniques [129].

Furthermore, cloud computing is another area that is receiving a lot of attention. Figure 3 illustrates the application of cloud computing in seizure prediction which can be employed in hospitals. The EEG signals obtained from the patients will be sent from the hospital to the cloud where the deep learning model is kept for prediction. The outcome of the diagnosis from the deep

learning model will be sent back to the hospital server. The hospital will contact the patient if there is any risk of seizure occurrence.

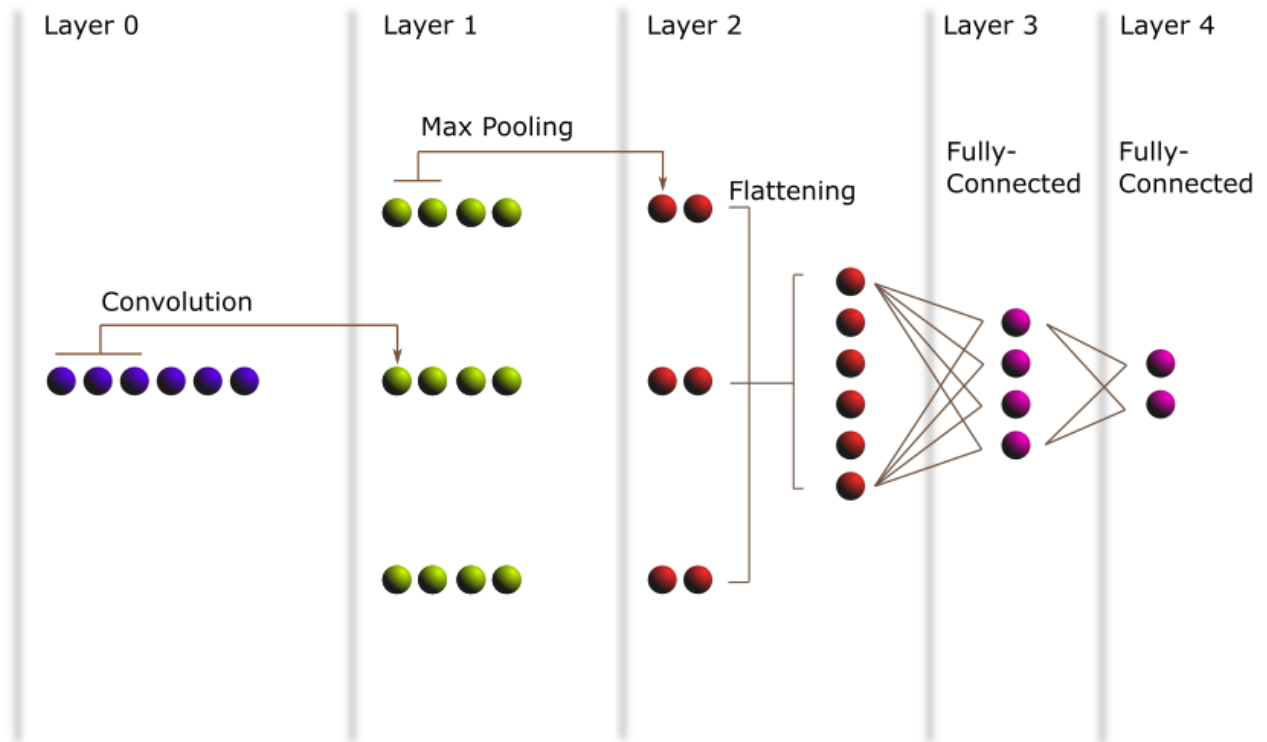


Figure 2 A simple CNN architecture.

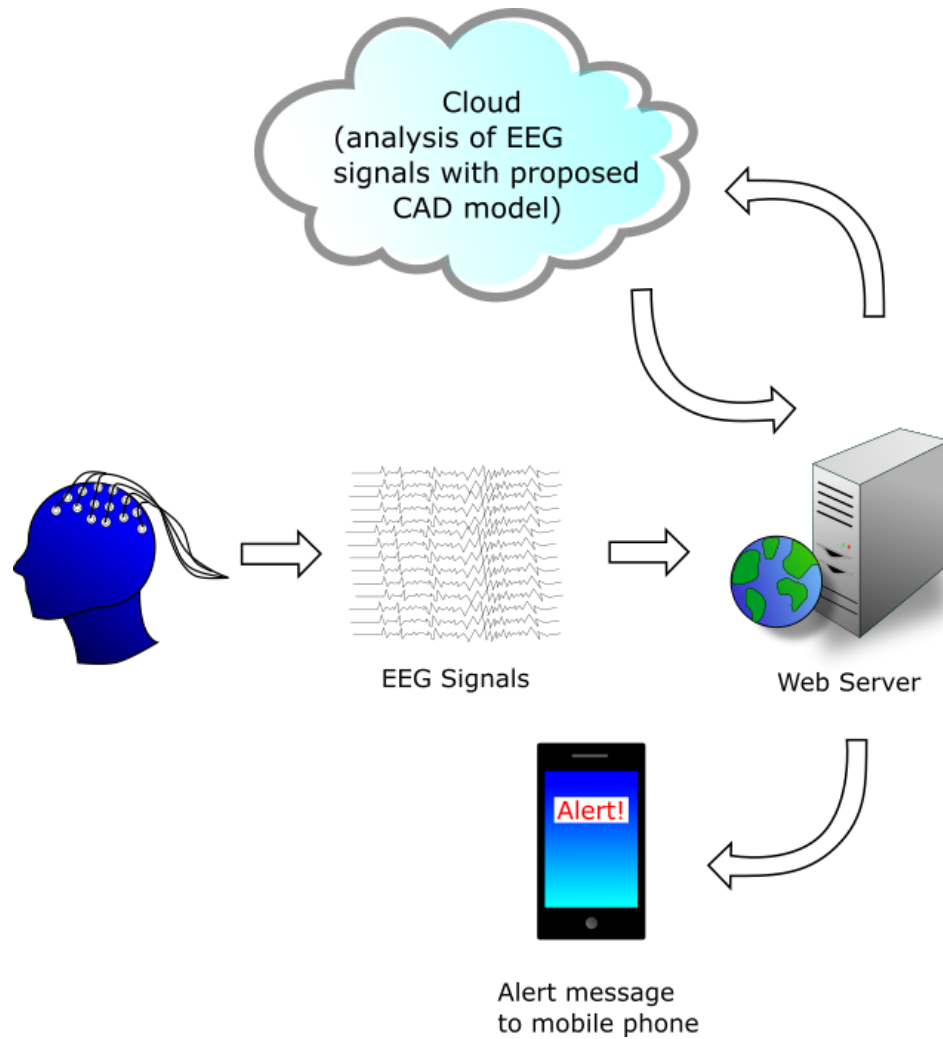


Figure 3 An illustration of a cloud computing process.

6. FINAL REMARKS

An automated accurate prediction of seizure will significantly improve the quality of life of patients reduce the burden on the caregivers. In this paper, the authors reviewed different seizure prediction algorithms, challenges faced, analysis methods and state-of-art techniques used. Future research shall focus on improving the accuracy and reliability of the seizure prediction models. Deep machine learning techniques, the same technology used in smart phones and autonomous cars, will play a role in this regard. Accurate and reliable prediction of seizure has the potential to revolutionize the treatment of epilepsy.

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